Introduction to MDPs & RL Core Concepts

Introduction to Reinforcement Learning

- **Core Concept:** RL is based on an agent learning by trial and error. Unlike supervised learning, the agent is not explicitly told what the correct action is; it must discover which actions yield the most reward by interacting with the world.
- **Reward Hypothesis:** All goals can be described by the maximization of expected total reward. If we want an agent to achieve a specific goal, we must provide rewards to it in such a way that maximizing those rewards corresponds perfectly with achieving the goal.
- **Delayed Rewards:** Rewards may be delayed, meaning an action taken now can result in positive or negative rewards much later in the future (after multiple steps in a trajectory). This introduces the challenge of *credit assignment*, determining which past actions were responsible for a future outcome.

The Agent-Environment Interaction

Environment: The world that the agent lives in, interacts with, and learns from. The environment defines the rules of the universe, including state transitions and reward logic.

The Loop: The interaction unfolds in a sequence of discrete time steps, $t = 0, 1, 2, 3, \ldots$ At each step t:

- **Agent:** Executes an action A_t, receives an observation O_t, and receives a scalar reward R_t.
- **Environment:** Receives the action A_t, updates its internal state, emits the next observation O_{t+1}, and emits a scalar reward R_{t+1}.

In summary: The agent sees a state $\rightarrow$ takes an action $\rightarrow$ receives a reward signal and the next state from the environment.

History, State, and Observability

History and State

- **History (H_t):** The complete sequence of observations, actions, and rewards up to time t:
$$H_t = O_1, R_1, A_1, \ldots, A_{t-1}, O_t, R_t$$
What happens next depends entirely on the history.

- **State (S_t):** The concise information used to determine what happens next. Formally, state is a function of the history: $S_t = f(H_t)$. It is a complete description of the world, whereas an **Observation (O_t)** is only a partial description.
- **Trajectories (τ):** A specific sequence of states, actions, and rewards over an episode.

Types of State

- **Environment State (S_t^e):** The environment's private representation. It contains whatever data the environment uses to pick the next observation/reward. It is not usually visible to the agent and often contains irrelevant or redundant information.
- **Agent State (S_t^a):** The agent's internal representation. It contains whatever information the agent uses to pick the next action. This is the information used and updated by RL algorithms.
- **Information State (Markov State):** Contains all useful information from the history. A state S_t is Markov if and only if it follows the Markov Property:

  $$P[S_{t+1} \mid S_t] = P[S_{t+1} \mid S_1, \dots, S_t]$$

  "The future is independent of the past given the present." Once the Markov state is known, the history may be thrown away.

Observability

- **Fully Observable Environments:** The agent directly observes the complete environment state ($O_t = S_t^a = S_t^e$). This is formally modeled as a **Markov Decision Process (MDP)**.
- **Partially Observable Environments:** The agent indirectly observes the environment (e.g., a robot with limited camera vision). Here, $S_t^a \neq S_t^e$. Formally, this is a **Partially Observable Markov Decision Process (POMDP)**. The agent must construct its own state representation, perhaps using beliefs or recurrent neural networks.

Major Components of an RL Agent

An RL agent typically includes one or more of the following key elements, operating within an **Action Space** ($\mathcal{A}$), which is the set of all valid actions available to the agent.

1. Policy (π)

- The agent's behavior function, serving as a map from the perceived states of the environment to the actions to be taken. It is the core of the RL agent.
- **Deterministic Policy:** $a = \pi(s)$. Maps a state exactly to one specific action.

- **Stochastic Policy:** $\pi(a \mid s) = P[A_t = a \mid S_t = s]$. Outputs a probability distribution over actions, which is highly useful for exploration.

2. Reward (R) and Return (G)

- **Reward Signal:** Defines the immediate, short-term goal of the RL problem.
- **Return (G_t):** The total expected cumulative reward from time step t.
- **Discount Factor ($\gamma \in [0,1]$):** A mathematical factor used to ensure the sum of future rewards converges in infinite-horizon problems, and to represent uncertainty about the future. The discounted return is defined as:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

  If $\gamma = 0$, the agent is myopic (only cares about immediate reward). As γ approaches 1, the agent becomes farsighted.

3. Value Functions

- Functions that predict the expected return to evaluate the goodness or badness of situations, helping the agent select between actions.
- **State-Value Function ($V_\pi(s)$):** Measures how good it is to be in a specific state s under policy π.

$$V_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$$

- **Action-Value Function ($Q_\pi(s,a)$):** Measures how good it is to take a specific action a in state s under policy π.

$$Q_\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$$

- **Optimal Value Functions (V^*, Q^*):** Represent the maximum possible expected return an agent can achieve across all possible policies.

4. Model (Optional)

- The agent's internal representation used to predict what the environment will do next. Agents that use models are called **Model-Based** (using planning), while those that don't are **Model-Free** (learning directly from experience).
- **Transitions (Dynamics):** Predicts the next state: $\mathcal{P}^a_{ss'} = P[S_{t+1} = s' \mid S_t = s, A_t = a]$.
- **Rewards:** Predicts the next immediate reward: $\mathcal{R}^a_s = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$.

Exploration vs. Exploitation

This is the fundamental trade-off the agent faces while learning:

- **Exploration:** Trying out new, untested actions to gather more information about the environment. This helps discover better long-term strategies.
- **Exploitation:** Using known information to choose the action that is currently expected to yield the highest reward to maximize short-term gains.
- **Common Strategies:** A standard approach to balance this is the **ϵ-greedy** strategy, where the agent exploits the best known action with probability $1 - \epsilon$, and explores a random action with probability ϵ.